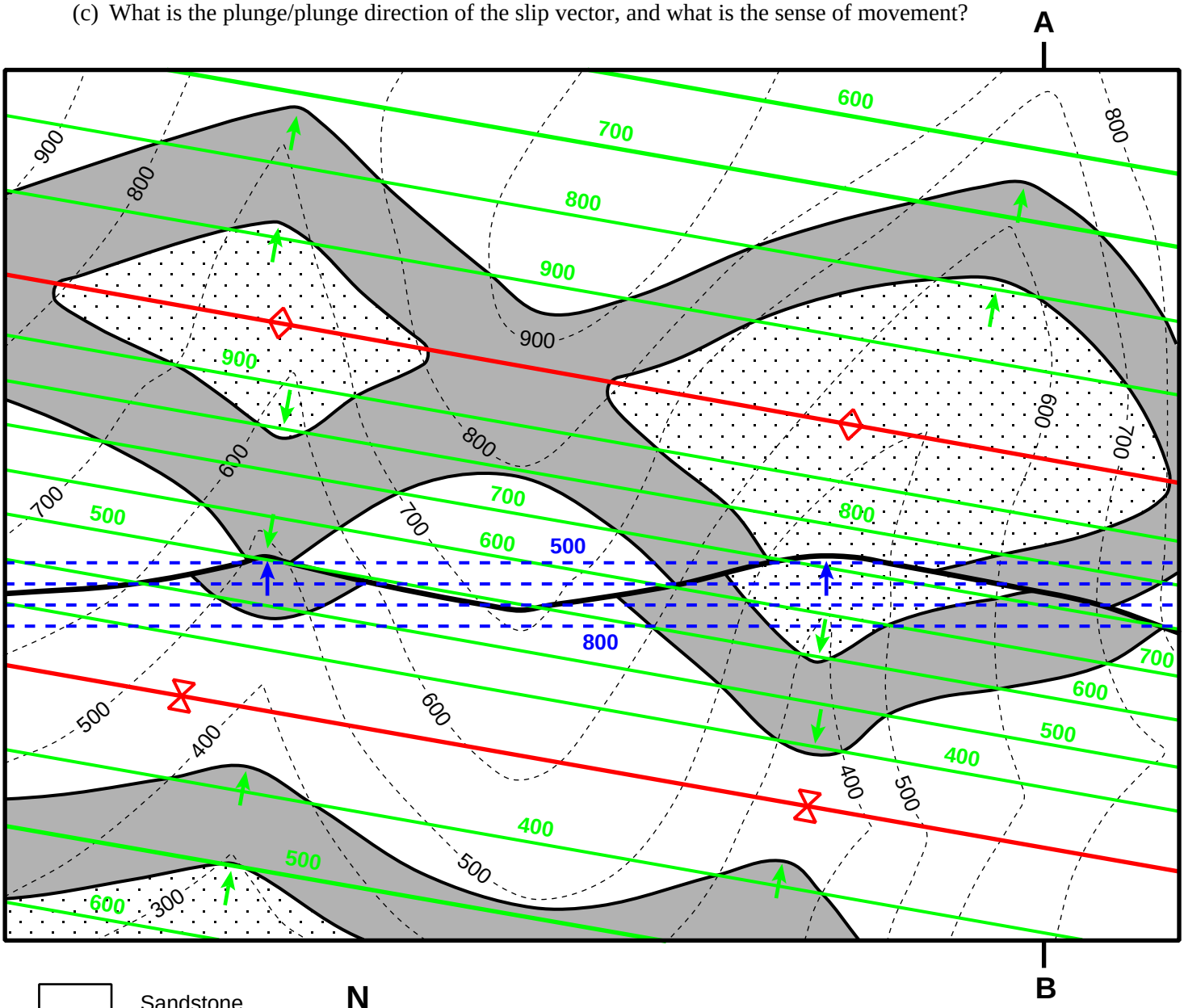
FOLDS AND FAULTS EXERCISE

1. By drawing stratum contours on the contact between the sandstone and the mudstone, determine the geometry of the folds shown on the map below. (HINT: use the V-ing of the contacts in the valleys to determine the approximate orientation of the stratum contours before trying to draw them.)
   (a) Determine the strike/dip of each limb of the folds.
   (b) Draw on the axial traces of the folds, and label them as antiform or synform.
   (c) What is the plunge/plunge direction of the fold axis?

2. The fault shown on the map is a pure dip-slip fault.
   (a) Determine the strike/dip of the fault plane.
   (b) By drawing a vertical cross-section along line A-B, determine the magnitude of the slip vector.
   (c) What is the plunge/plunge direction of the slip vector, and what is the sense of movement?



Sandstone

Mudstone

Grit

N

Scale 1:40000 ; contours in metres

↙ approximate dip direction given by veeing

Fold geometry

Limbs : 100/14N and 100/19S

Fold axis : 0/100

Bisector : 100/88N (≈ axial plane)

Interlimb angle : 147°

Fault geometry

Orientation : 090/34N

Slip vector : 34/000 ; 120 m (normal)